

Aiden Aysun Portfolio

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Media (Website) link

GitHub link

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About This Maker Portfolio

I am deeply passionate about complex problem-solving and the engineering design process. Over the course of my STEM career, I have undertaken numerous projects. Through these projects, I have been able to independently pursue STEM topics that I am truly interested in, and I have become well-versed in the engineering design process. In this portfolio I highlight several key undertakings that best reflect my experience and growth. Through these initiatives I have not only honed my technical capabilities, but developed an enduring enthusiasm for scientific inquiry and for translating conceptual ideas into engineered solutions. I plan to carry this mindset into future academic and professional pursuits.

1. Robotics

My introduction to STEM began via robotics in middle school, and by high school I committed four years as a member of our FIRST Tech Challenge (FTC) robotics team. I served as CAD Lead for two years and Hardware Lead for one year, actively contributing to the design, modeling, and build processes. I modeled the robot's core mechanisms in Autodesk Fusion 360, integrating principles of mechanical design, kinematics, and manufacturing constraints to optimize system performance. Additionally, I designed and assembled multiple hardware subsystems and developed autonomous control programs to implement precise, real-time robot behaviors.

1.1 CAD Projects

Intake and Claw Design

One challenge of the 2023-2024 FTC season was to collect pyramid-shaped game elements that were arranged in random orientations.

In light of this, our robot began with a passive intake system, but the design exhibited limitations: weak grip when clipping, and inconsistency when acquiring pieces from arbitrary orientations. To resolve these issues, I designed a custom claw mechanism: a gearbox-driven actuator equipped with induced-fit “pyramid-shaped” fingers optimized for form-fit contact. I then designed a servo-driven mechanism that provided three degrees of freedom (rotation about the x , y , and z axes) to orient the claw appropriately. This enhanced geometry not only improved contact reliability but also increased scoring efficiency, and was integrated into the final robot version.

Flywheel Shooter and Sorting Mechanism

For a subsequent game challenge - shooting plastic ball artifacts via a high chute - I designed an advanced shooter system featuring:

- A flywheel/inertia wheel to launch the ball, balancing speed and accuracy by maintaining contact for a controlled acceleration.
- A “flapper” mechanism to adjust the launch angle via a linkage and an encoder-equipped Axon servo.
- A compact “spindexer” (rotating indexer) to store and transition the balls from the intake to the shooter while satisfying the 18 inch size constraint.

Flywheel Design

Determining flywheel mass m and radius r required modeling rotational inertia I and launch angular velocity ω . For a solid disk approximation:

$$I = \frac{1}{2}mr^2$$

The launch linear speed v of the ball is related by $v = r\omega$. The stored rotational kinetic energy is:

$$E = \frac{1}{2}I\omega^2 = \frac{1}{4}mr^2\omega^2$$

Matching this energy to the energy required to accelerate the ball (mass m_b and target speed v_b):

$$\frac{1}{4}mr^2\omega^2 \approx \frac{1}{2}m_bv_b^2$$

This relation guided selection of m and r satisfying geometric and performance constraints. I verified the radius of the flywheel by modeling the path of the ball along the curved flywheel shooter ramp in Fusion360. Additionally, the modular shooter design incorporated mounting points for attachment to a “lazy-susan” turret that allows for more accurate shooting.

Spindexer Design

Rather than a linear sorting conveyor, I designed a circular spindexer (12 inch diameter) using three custom walls and an integrated “kicker” that both indexed and fed balls into the shooter. I reserved space for future color sensors to enable automatic sorting. Using

Fusion360's motion study, I visualized the full assembly - intake, spindexter, shooter, flapper, and turret - to verify kinematic clearances and mechanical stability.

Relevant Media (Website)

1.2 Torque and Motor Selection Calculations

Beyond CAD modeling, I contributed to hardware analysis. When selecting linear actuator motors, I applied physics and calculus concepts to compute the minimum required torque rather than relying on trial-and-error.

Given subsystem mass m and required acceleration a :

$$F = ma$$

If the actuator uses a lead screw of effective radius r_s , the torque τ requirement is:

$$\tau = Fr_s = mar_s$$

Accounting for mechanical efficiency $\eta < 1$:

$$\tau_{\text{motor}} \geq \frac{\tau}{\eta}$$

I computed efficiency curves and matched gearbox ratios to find the optimal torque-speed balance motor of 435 RPM, minimizing losses and heat generation.

Relevant Media (Website)

1.3 Software and Control Systems

Autonomous Pathing

Previously, we used a timed position drive in our autonomous navigation system. This autonomous program had rigid waypoint motion (stop-and-go) that was slow and inefficient. To address this, I developed a smoother path system by logging the global pose (x, y, θ) of the robot using odometry and IMU data. I then took this logged data in a plotter to generate optimized paths. As a result, implementing a continuous drive-to-position controller

significantly improved runtime efficiency and precision. This system has been used in our 2023–2025 FTC seasons.

Menu Interface

In the 2023 FTC season’s game, it was critical to coordinate our autonomous with our alliance’s autonomous due to the shared space and risk of collision. Coordinating with our autonomous with our alliance partner required maintaining a set of configurable autonomous parameters such as start and truss delays, park location, and specific autonomous program. I decided that a Menu Interface would be a great way to organize the autonomous parameters and allow for the most precise autonomous. The menu system utilizes a finite state machine and cycles through six menu entries in a bounded wrap-around fashion.

```
if (nextEntry) { initMenuSelected++; if (initMenuSelected > initMenuMax) initMenuSelected = 1; }
if (prevEntry) { initMenuSelected--; if (initMenuSelected < 1) initMenuSelected = initMenuMax; }
```

The “initMenuSelected” variable acts as a pointer to the current menu entry it is modifying. This allows for real-time configuration of autonomous parameters. For example, if we want to add a start delay to our autonomous to prevent a collision, or alter the color detected if our vision pipeline is having trouble, this Menu Interface allows us to make these adjustments.

To cycle between each menu entry and the specific menu entry value, I used modular indexing and a series of switch cases so that the menu interface performs accurate context-sensitive control behavior.

```
switch(initMenuSelected) {
  case 1:
    if(nextValue) { if(startDelaySec < 9) startDelaySec++; }
    if(prevValue) { if(startDelaySec > 0) startDelaySec--; }
    break;

  ~~~~
  case 6 : // HOW MANY PIXELS DO WE SCORE FROM 5-STACK?
    if( nextValue ) {if (fiveStackCycles < 1) {fiveStackCycles++;}} // next
    if( prevValue ) {if (fiveStackCycles > 0) {fiveStackCycles--;}} // prev
    break;
}
```

This menu interface proved invaluable in optimizing our autonomous routine to coordinate seamlessly with our alliance partners and adapt to varying field conditions. Its modular

design also allowed us to reuse and refine the same system for the 2024–2025 season. By enabling rapid pre-match configuration and improved team coordination, the interface provided a competitive advantage that helped our team advance to the FTC State Championship. Refer to the links below for relevant media and the complete source code on GitHub.

Relevant Media (Website)

Github Project

2. Spectral Sensing Research

Varying illumination conditions can significantly degrade the accuracy of color classification systems. My triplet brothers and I experienced this firsthand when our team’s autonomous color detection algorithm failed under inconsistent lighting during the FTC League Championship - costing us advancement to the State Tournament. Recognizing the real-world importance of robust color sensing, we set out to design a more reliable detection system.

Inspired by the precision of astronomical spectrometers - used to identify exoplanets despite highly variable lighting environments - we implemented a 10-channel Adafruit AS7341 spectral sensor interfaced with an Arduino Nano to capture calibrated wavelength-intensity data. Through extensive experimentation, we found that a machine learning-based Random Forest classifier achieved substantially higher accuracy than a traditional Cosine Similarity approach under diverse illumination conditions. This project deepened my understanding of spectral sensing, data preprocessing, Arduino, sensor communication, machine learning (Random Forest), color theory, and calibration methodology. While each of us contributed equally, my primary responsibility involved integrating the CAD and hardware systems - ensuring precise alignment and secure interfacing between the Arduino Nano and spectral sensor to enable consistent spectral acquisition. A detailed research paper outlining our methodology, analysis, and experimental results, along with our GitHub repository documenting our research, is available below.

Full Research Paper (PDF) on Website

Github Project

3. Assistive Notes Processor

When I observed students struggling to read lecture notes due to impaired eyesight, I created an assistive software pipeline. This project implements a complete pipeline that transforms scanned or photographed handwritten notes into high-contrast, more readable PDFs using Python-based image processing and file conversion methods. The system is divided into three main modules:

1. PDF-to-Image Conversion
2. Image Preprocessing and Binarization
3. Image-to-PDF Reconstruction

This workflow allows handwritten notes with uneven lighting, low contrast, or background noise (e.g., notebook lines or shadows) to be processed into clean documents optimized for readability.

1. PDF-to-Image Conversion

The conversion pipeline begins by using the `pdf2image` library with the Poppler backend to rasterize each page of a PDF into high-resolution `.png` images:

```
from pdf2image import convert_from_path
pages = convert_from_path(pdf_path=pdf_path, poppler_path=poppler_path)
```

Each page is stored as an image in a working directory. The conversion from vector-based PDF to pixel-based image allows direct manipulation of pixel intensity values.

Mathematically, this step performs a sampling of the continuous PDF representation $P(x, y)$ into a discrete image matrix:

$$I[i, j] = P\left(\frac{i}{R_x}, \frac{j}{R_y}\right)$$

where R_x and R_y are the image resolutions in pixels per inch (PPI).

2. Image Processing and Color Conversion

The next stage processes each image to increase legibility using grayscale conversion and binary thresholding. The program utilizes the Python Imaging Library (PIL) to iterate over every pixel and apply the luminance-based grayscale equation:

$$Y = 0.299R + 0.587G + 0.114B$$

This formula models human visual perception of brightness. After computing Y , a custom thresholding function is applied to binarize the image:

```
# Apply formula of grayscale:
newVal = (0.299*r + 0.587*g + 0.114*b)
if(r>240 and g > 240 and b > 240):
    newVal = 0
else:
    newVal = 255

# setting the pixel value.
pixel_map[i, j] = (int(newVal), int(newVal), int(newVal))
```

Thus, all light areas (paper background) are mapped to black (0), while darker strokes (writing) are mapped to white (255)—effectively producing a *white-text-on-black-background* image that maximizes contrast.

This resembles a global binary thresholding technique:

$$B(x, y) = \begin{cases} 0, & \text{if } Y(x, y) > T \\ 255, & \text{otherwise} \end{cases}$$

where $T \approx 240$ in this implementation.

3. Image-to-PDF Reconstruction

Once processed, all images are reassembled into a single PDF using the `img2pdf` library:

```
with open("Output.pdf", "wb") as file:
    file.write(img2pdf.convert(
        [i for i in os.listdir(folder) if i.endswith(".png")]
    ))
```

This step concatenates all binarized `.png` images into a vector-wrapped raster PDF, ensuring high quality without recompression artifacts. Unlike PIL's built-in PDF export, `img2pdf` preserves resolution and color depth precisely, using a lossless pipeline.

This system effectively transforms imperfect handwritten documents into professional-grade, print- and archive-ready digital files.

Relevant Media (Website)

Github Project

4. YouTube Rewind Controller Chrome Extension (Published)

Overview

This project implements a Chrome extension that adds low-latency rewind functionality to YouTube videos. By holding the `R` key, the user can continuously rewind playback in small increments. The system leverages the Chrome Extensions Manifest V3 framework and consists of a background service, a content script, and a dynamically injected overlay for user feedback.

System Architecture

The extension follows a modular, event-driven design:

- **Manifest Configuration:** Defines host permissions for `youtube.com` and injects scripts upon page load.
- **Background Script:** Monitors browser tabs and sends messages to the content script when a new video loads.
- **Content Script:** Intercepts keyboard input, modifies playback state, and handles DOM injection.
- **Overlay UI:** Displays an active-state indicator ( `R`) when rewinding.

Rewind Algorithm

The content script locates the active HTML5 `<video>` element and modifies its current playback time:

$$t_{new} = \max(0, t_{current} - \Delta t)$$

with $\Delta t = 0.2$ s, updated every 100ms via a timer loop. This discrete-time rewind model mimics continuous negative playback by manipulating frame indices directly.

```
if (e.key === 'r' || e.key === 'R') {
  if (!rewindInterval) {
    rewindInterval = setInterval(() => {
      showOverlay();
      const video = getYouTubeVideo();
      if (video) {
        //video.playbackRate = -1;
        video.currentTime = Math.max(0, video.currentTime - 0.2);
      }
    }, 100);
  }
}
```

Releasing the key cancels the interval and hides the overlay:

```
document.addEventListener('keyup', (e) => {
  if (e.key === 'r' || e.key === 'R') {
    clearInterval(rewindInterval);
    rewindInterval = false;
    hideOverlay();
  }
});
```

Communication and Control Flow

The background script detects tab updates:

```
chrome.tabs.onUpdated.addListener((tabId, tab) => {
  if(tab.url&&tab.url.includes("youtube.com/watch") ){
    const queryParameters = tab.url.split("?")[1];
    const urlParameters = new URLSearchParams(queryParameters);

    chrome.tabs.sendMessage(tabId, {
      type : "NEW",
      videoId : urlParameters.get("v"),
    });
  }
})
```

This message-driven architecture decouples page monitoring from playback logic, ensuring efficient asynchronous execution.

I developed this Chrome extension using JavaScript, HTML, and CSS. By interacting with the Chrome Extensions API (Manifest V3) and the HTMLMediaElement interface, my Chrome Extension enables precise manual rewind functionality to YouTube videos. Its event-driven architecture integrates real-time DOM manipulation, message passing, and timer-based state updates to achieve low-latency performance. The system enables continuous playback reversal through intuitive keyboard control, improving video navigation for analysis and learning tasks. This project showcases the practical application of browser automation, media API integration, and efficient client-side event handling.

Relevant Media (Website)

Chrome Web Store

Github Project

Conclusion

Each project in this portfolio reflects my growth across mechanical design, CAD modeling, control systems, applied physics, and software development. I aim to continue exploring interdisciplinary engineering, particularly in aerospace and mechanical systems, where innovation and analytical rigor converge.